

From Time to Frequency: A Hilbert Space Perspective on DTFT and Parseval's Identity

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Motivation

- This project began with a doubt: when moving from a countable basis (time-domain sequence in l^2) to an uncountable basis (frequency-domain DTFT), wouldn't norm/inner product preservation fail since the bases 'don't match'?
- In finite dimensions, we can represent transformations by a fat or tall matrix with orthonormal rows or columns, which preserves inner products. But with an uncountable basis, we cannot enumerate such a matrix.
- This motivated us to step up into Hilbert space theory, where Parseval's theorem guarantees that the DTFT is in fact an isometry — preserving energy and inner products across domains.

What Is a Hilbert Space?

A Hilbert space is a complete inner product space.

That means

- It has an inner product $\langle x, y \rangle$
- Induces a norm $\|x\| = \sqrt{\langle x, x \rangle}$
- Is complete: all Cauchy sequences converge within the space

Examples

- $\mathbb{R}^n$ with dot product (finite-dimensional space)
- $\ell^2(\mathbb{Z})$: square-summable sequences
Most discrete-time signals we study in DSP belong here $\sum_{n=-\infty}^{\infty} \|x[n]\|^2 < \infty$
- $L^2([-\pi, \pi])$: space of square-integrable functions over frequency
DTFT representations $X(e^{j\omega})$ of DSP signals live in this space $\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 \frac{d\omega}{2\pi} < \infty$

What Makes Hilbert Spaces Powerful?

They generalize Euclidean geometry to infinite dimensions, but still preserve all key geometric ideas:

- Inner product $\implies$ lets us talk about angles, orthogonality
- Norm from inner product $\implies$ measures energy
- Orthogonal basis $\implies$ every signal is a “generalized linear combination”
- Projection theorem $\implies$ best approximation in a subspace still works
- Completeness $\implies$ limits of approximations don't “escape” the space

In DSP, this allows us to:

- Build signals from orthogonal basis functions (like $e^{-j\omega n}$)
- Interpret energy and correlations through inner products
- Apply transforms (like DTFT) that respect these structures

DTFT as a Map Between Hilbert Spaces

We now treat the DTFT not just as a formula, but as a linear transformation:

$$\mathcal{F} : \ell^2(\mathbb{Z}) \mapsto L^2([-\pi, \pi]):$$

This map is:

- Linear: $\mathcal{F}(ax + by) = a\mathcal{F}(x) + b\mathcal{F}(y)$
- Norm-preserving (Isometry):
$$\|x\|_{\ell^2}^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega = \|X\|_{L^2}^2$$
- Inner-product preserving: $\langle x, y \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) \overline{Y(e^{j\omega})} d\omega$

In other Words:

- The DTFT respects the Hilbert space structure
- It is an isometry between time-domain and frequency-domain signal spaces.

DTFT Operator setup

- Consider the subspace $c_{00} \subset \ell^2$, where c_{00} is dense subspace of finite-support sequences
- Now consider the DTFT operator(normalized):
$$(T_0 x)(\omega) = \frac{1}{\sqrt{2\pi}} \sum_n x[n] e^{-j\omega n}$$
- Here basis functions are
$$\phi_n = \frac{1}{\sqrt{2\pi}} e^{-j\omega n}$$
- With target space being $L^2([-\pi, \pi])$ inner product:
$$\langle F, G \rangle = \int_{-\pi}^{\pi} F(\omega) \overline{G(\omega)} d\omega$$

Proof that DTFT is isometry and is infact unitary

- We know $\forall x, y \in c_{00} : \langle (T_0 x)(\omega), (T_0 y)(\omega) \rangle = \int_{-\pi}^{\pi} (T_0 x)(\omega) \overline{(T_0 y)(\omega)} d\omega$

$$\begin{aligned} \Rightarrow \langle (T_0 x)(\omega), (T_0 y)(\omega) \rangle &= \int_{-\pi}^{\pi} \frac{1}{\sqrt{2\pi}} \sum_n x[n] e^{-j\omega n} \overline{\frac{1}{\sqrt{2\pi}} \sum_n y[n] e^{-j\omega n}} d\omega \\ &= \int_{-\pi}^{\pi} \sum_n \sum_m x[n] \overline{y[m]} \left(\frac{1}{\sqrt{2\pi}} e^{-j\omega n} \right) \left(\frac{1}{\sqrt{2\pi}} \overline{e^{-j\omega m}} \right) d\omega \\ &= \sum_n \sum_m x[n] \overline{y[m]} \int_{-\pi}^{\pi} \left(\frac{1}{\sqrt{2\pi}} e^{-j\omega n} \right) \left(\frac{1}{\sqrt{2\pi}} \overline{e^{-j\omega m}} \right) d\omega \\ &= \sum_n \sum_m x[n] \overline{y[m]} \langle \phi_m, \phi_n \rangle \\ &= \sum_n \sum_m x[n] \overline{y[m]} \delta_{mn} \end{aligned}$$

- Note: Since sequences are of finite support, the sums are finite and the interchange of summations/integral is always valid.

Orthogonality and Conclusion

- Where $\langle \phi_m, \phi_n \rangle = \delta_{mn}$ arises because of orthogonality
- Since

$$\delta_{mn} = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{if } m \neq n \end{cases}$$

The double summation collapses to single summation.

-

$$\begin{aligned} \langle (T_0 x)(\omega), (T_0 y)(\omega) \rangle &= \sum_n x[n] \overline{y[n]} \\ &= \langle x, y \rangle \end{aligned}$$

- If $x = y \implies \langle (T_0 x)(\omega), (T_0 x)(\omega) \rangle = \langle x, x \rangle$
- Hence T_0 is unitary

Extension to All ℓ^2

- c_{00} is dense in ℓ^2
- T_0 is isometry $\implies \|T_0\| = 1$

Theorem

(Bounded Linear Extension) [Kreyszig, Introductory Functional Analysis with Applications, Theorem 2.7-11].

Let $T : \mathbb{D}(T) \longrightarrow Y$ be a bounded linear operator with domain $\mathbb{D}(T)$ dense in a Normed space X and Y a Banach space Then T has a unique bounded linear extension $\tilde{T} X \longrightarrow Y$ satisfying:

$$\tilde{T}|_D = T \text{ and } \|\tilde{T}\| = \|T\|$$

- From this theorem, the DTFT operator T_0 (defined on c_{00}) extends uniquely to all ℓ^2 preserving boundedness, inner product and norm. Hence, the DTFT is well-defined and isometric (in fact unitary) on $\ell^2(Z)$

Why the Extension is Important in DSP

- In DSP, the DTFT is written as
$$X(e^{j\omega}) = \sum_{n=-\infty}^{n=\infty} x[n]e^{-j\omega n}$$
which converges absolutely only when $x \in \ell^1$
- Yet, many practical finite-energy signals belong only to ℓ^2 not in ℓ^1
(example: $x[n] = \frac{1}{n}, n \geq 1$)
- The extension theorem provides a rigorous foundation for this intuitive DSP practice: it defines the DTFT as an L^2 -limit, ensuring it remains well-defined and energy-preserving (isometric) for all ℓ^2 signals.

Numerical Verification — Why the Extension Matters

- Consider $x[n] = \frac{1}{n}$, $n \geq 1$.
- $x \in \ell^2$ but $x \notin \ell^1$.
- Its DTFT is $X(e^{j\omega}) = -\ln(1 - e^{-j\omega})$, which diverges at $\omega = 0$.
- Yet the total spectral energy remains finite:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega = \sum_{n=1}^{\infty} |x[n]|^2 = \frac{\pi^2}{6}.$$

Numerical Verification(Plots)

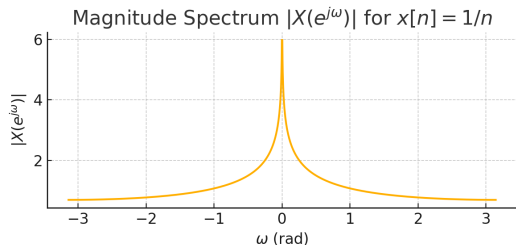


figure: Magnitude spectrum $|X(e^{j\omega})|$ diverges near $\omega = 0$.

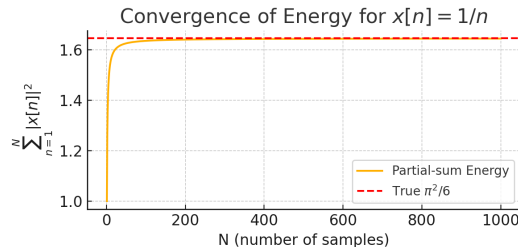


figure: Partial-sum energy converges to $\pi^2/6$, confirming finite total energy.

Takeaway: This example illustrates that while $x[n] = 1/n$ lacks a pointwise DTFT due to divergence, its transform exists as an L^2 -limit under the extended DTFT operator — validating the extension theorem.

Parseval's Identity —

From DSP Intuition to Hilbert-Space Rigor

- From the convolution orthogonality proof in DSP:

$$\sum_n x[n] \overline{y[n]} = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) \overline{Y(e^{j\omega})} d\omega$$

- For $x = y$:

$$\|x\|_{\ell^2}^2 = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

This establishes norm preservation — the core of Parseval's relation.

Interpretation

- The DSP derivation already displays the isometric property of the DTFT in form.
- The extension theorem later ensures this result is rigorously valid $\forall \ell^2$, including sequences not in ℓ^1 completing the proof of unitarity.

Key Takeaway

- Parseval's identity isn't a new result; it's the energy-preservation principle underlying DTFT unitarity.
- The DSP proof intuited it, and the Hilbert-space framework justifies it.

Beyond DTFT — From Cartesian to Radial Domains

- The DTFT arises because exponentials $e^{-j\omega n}$ form an orthonormal basis of $L^2([-\pi, \pi])$.
Each signal is expanded in these discrete frequency modes.
- In 2D polar coordinates, orthogonality changes form:
Angular : $e^{jm\theta}$
Radial: Bessel function $J_r(\cdot)$
- This leads to the Fourier–Bessel / Hankel Transform:
$$X(\rho) = \int_0^\infty x(r) J_r(\rho r) r dr$$

which defines a unitary (energy-preserving) mapping
$$\mathcal{H} : L^2([0, \infty), r dr) \longrightarrow L^2([0, \infty), \rho d\rho).$$
- Applications: used in radar Doppler / micro-Doppler analysis, wavefields, and circularly symmetric signal processing.

Key Insight — What This Reveals

- The familiar Parseval identity of DSP is not just an algebraic energy relation — it expresses a deep Hilbert-space isometry: a transform that preserves inner products and norms.
- By viewing the DTFT as a unitary operator: $\mathcal{F} : \ell^2(\mathbb{Z}) \mapsto L^2([-\pi, \pi])$, we see that Fourier analysis is a geometry-preserving mapping between two function spaces.
- Extending this idea to radial domains, the Hankel transform
$$X(\rho) = \int_0^\infty x(r) J_r(\rho r) r dr$$
is also unitary — energy and orthogonality are preserved, though its “basis” functions form a continuous rather than discrete system.
- These unitary mappings are the mathematical backbone of: signal stability and energy conservation in DSP, and modal decompositions in radar and wavefield analysis.

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- Contributors implemented supporting numerical simulations and reviewed the final report
- Based on ideas originating in EE603 class discussions and mail correspondences

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